

[SEC-OPS-2026-04] [Critical multi-stage alert triggered by finance user - Confirmed False Positive](#)

Created: 13/Mar/26 Updated: 14/Mar/26 Resolved: 13/Mar/26

|                   |                                |
|-------------------|--------------------------------|
| Status:           | Done                           |
| Project:          | SEC-OPS                        |
| Components:       | SIEM (WAZUH),Endpoint Security |
| Affects versions: | None                           |
| Fix versions:     | None                           |

|                     |                                                                        |           |                            |
|---------------------|------------------------------------------------------------------------|-----------|----------------------------|
| Type:               | Incident                                                               | Priority: | Highest                    |
| Reporter:           | <a href="#">Sai Prasad</a>                                             | Assignee: | <a href="#">Sai Prasad</a> |
| Resolution:         | False Positive                                                         | Votes:    | 0                          |
| Labels:             | false-positive, finance, siem-rule-tuning, ssh, sudo, wazuh, whitelist |           |                            |
| Remaining Estimate: | 0h                                                                     |           |                            |
| Time Spent:         | 1h 25m                                                                 |           |                            |
| Original estimate:  | 2h                                                                     |           |                            |

|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attachments: | <div><div>ssh-keys.png</div><div>crontab-check.png</div><div>New systemd services.png</div><div>lateral-movement.png</div><div>kernel-version.png</div><div>log-integrity.png</div><div>wazuh-alert-100101.png</div><div>Screenshot 2026-03-13 123707.png</div><div>stat-backup.png</div><div>wazuh-alert-100200.png</div><div>stat-original.png</div><div>hash-comparison.png</div><div>ssh-failures.png</div><div>sudo-pattern.png</div><div>wazuh-alert-100400.png</div><div>network-listeners.png</div><div>exfiltration-check.png</div><div>wazuh-alert-100300.png</div><div>bash-history.png</div><div>wazuh-alert-100100.png</div><div>finance-access-pattern.png</div><div>wazuh-alert-100999.png</div><div>active-sessions.png</div><div>backup-structure-2026.png</div><div>backup-pattern.png</div><div>network-connections.png</div><div>exfiltration-check continuation.png</div><div>ssh-process-id.png</div><div>virusotal-ip.png</div><div>backup-structure-2025.png</div><div>auditd-status.png</div><div>ssh-success.png</div></div> |
| Rank:        | 0 i0004f:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

Description

On March 13, 2026 at 11:03 IST, Wazuh SIEM triggered Rule 100999 (CRITICAL) for user sp on Agent3-ubuntu (192.168.29.90). The correlation rule detected a sequence of events within 300 seconds: two SSH authentication failures followed by success, access to Tier-0 finance directory, sudo privilege escalation, and backup file creation in /root/finance\_backups/.

Key Findings:

- Two SSH failures were typographical errors - all three attempts shared same process ID [8064], confirming they occurred within same session

- User accessed Finance/January directory as part of routine month-end verification
- Permission denied on sensitive payroll/audit files - security controls working
- User escalated via sudo -s (authorized), created backup directory, copied three files, applied restrictive permissions (700/600)
- Historical baseline analysis (7 days) shows daily finance access and backup operations 3-4 times weekly
- Backup structure spans 2+ years with monthly directories - established practice
- SHA256 hashes identical between original and backup files - NO DATA MODIFICATION
- NO exfiltration - zero file transfer commands, no outbound connections
- NO persistence - no unauthorized cron jobs, no new SSH keys, no new systemd services
- NO lateral movement - all connections from 192.168.29.50 only
- NO log tampering - auth.log rotation normal, auditd running continuously
- NO threat intelligence hits - all artifacts legitimate

#### **Actions Taken:**

- Investigation paused at 11:26 to prioritize Windows critical alert (WIN-ENDPOINT-42)
- Investigation resumed 12:17
- User contacted at 12:21 - confirmed legitimate activity
- No isolation required

**Verdict:** False Positive - Legitimate Administrative Activity

**Impact:** None (No data loss, no modification, no exfiltration)

**Closure:** 35 minutes active investigation time (1 hour 25 minutes total with pause)

#### **Comments**

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Logged into Wazuh and observed Rule 100999 triggered for user sp on Agent3-ubuntu. The correlation rule fired due to a sequence within 300 seconds: two SSH failures followed by success, finance directory access, sudo escalation, and backup creation. Source IP 192.168.29.50 is internal. Timestamp 11:01-11:06 is business hours. Beginning investigation now.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

SSH authentication analysis complete. Examined last, lastb, and auth.log. All three attempts share the same SSH process ID [8064], confirming they occurred within the same session. Two failures followed by success from the same source IP within 10 seconds is definitive evidence of typographical error. No evidence of brute force or credential compromise. This does not appear to be an SSH attack - just a typo error. Pausing investigation to prioritize other tickets while awaiting additional context.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Resuming investigation now. Continuing with command history analysis.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Command history analysis complete. User navigated to Finance/January directory, attempted to read payroll and audit files (permission denied - security controls working), successfully

read revenue file, escalated via sudo -s, created organized backup directory, copied three files to /root/finance\_backups/jan\_2026/, applied restrictive permissions (700/600), and verified copy. No file transfers, no modifications, no deletions. Sequence matches authorized workflow.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Historical baseline analysis complete. Reviewed command history from past 7 days. User accesses finance directories daily, performs backup operations 3-4 times weekly, and uses sudo -s consistently. Today's activity matches established behavioral pattern. This appears to be routine month-end financial data verification.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Examined backup directory structure. Backup structure spans 2024-2026 with monthly directories and consistent naming. Executive Payroll files present for every month. This is a long-standing backup practice (2+ years), not anomalous activity.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

File integrity verification complete. Original file modify time is March 12 (yesterday) - unchanged during today's session. Backup timestamps match copy operation at 11:05:47. SHA256 hashes are identical between original and backup files. Cryptographic proof of exact copy with no modification or tampering.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Exfiltration check complete. No file transfer commands in today's history (only SCP from Jan 15 for SSH key setup). No outbound connections beyond current SSH session. No connections to external IPs. No data exfiltration detected.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Persistence check complete. No unauthorized cron jobs, no new SSH keys, no unusual network listeners, no new systemd services. No backdoors or persistence mechanisms detected.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Lateral movement check complete. All accepted SSH connections from 192.168.29.50 only. Known\_hosts file last modified Jan 15 with no new hosts added today. No lateral movement detected.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Log integrity check complete. Authentication logs show normal daily rotation at 00:00 UTC with no gaps or truncation. Auditd has been running continuously since Mar 10 with no restarts during the incident window. No log tampering or defense evasion detected.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Threat intelligence correlation complete. Source IP 192.168.29.50 is private RFC1918 with no public reputation. File hash never submitted to VirusTotal. Username and hostname show 0 OTX pulses. Kernel version 6.17.0-14 is one patch behind current. No threat intelligence hits. Kernel patch ticket INFRA-PATCH-2026-0213 created.

Comment by [Sai Prasad](#) [ 13/Mar/26 ]

Root cause analysis complete. Immediate cause: two SSH typos combined with legitimate finance activity triggered correlation rule. Contributing factors: rule lacks whitelist, no change ticket for routine access, no behavioral context. Investigation definitively proves false positive based on process ID evidence, 7-day baseline, 2+ year backup structure, and file integrity verification.

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